

cbse class 9 science notes chapter 3 atoms and molecules

Ancient Indian philosopher Maharishi Kanad proposed that matter is divisible until the tiniest particles, called "Parmanu." Similarly, Greek philosopher Democritus introduced the concept of indivisible particles, "atoms." These ideas were philosophical until the 18th century when scientists began distinguishing elements from compounds. Antoine Lavoisier established the foundation of modern chemistry with his laws of chemical combination.

Laws of Chemical Combination

1. **Law of conservation of mass:** Mass can neither be created nor destroyed in a chemical reaction.
2. **Law of Definite Proportions:** Elements are always present in fixed proportions by mass in a chemical compound.

Dalton's Atomic Theory Postulates

- All matter is made of tiny particles called atoms and are indivisible.
- Atoms of all elements can neither be created nor destroyed in chemical reactions.
- Atoms of an element are the same in mass and properties.
- Atoms of different elements possess different masses and different properties.

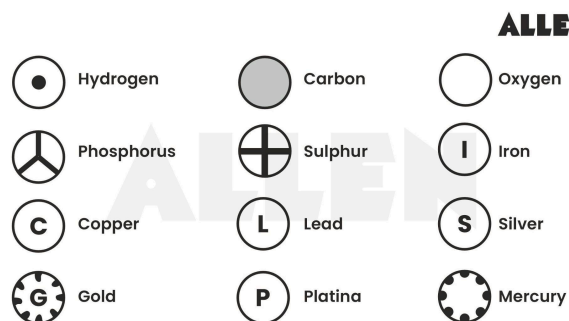
- Atoms combine in small whole-number ratios to form compounds.
- The number and types of atoms in a compound are constant.

Drawbacks of Dalton's Atomic Theory

- Atoms are divisible into subatomic particles (protons, neutrons, electrons).
- Dalton's theory doesn't explain isotopes, atoms of the same element with different masses.

Elements and Symbols

Dalton introduced a system of notation to represent elements, while Berzelius proposed the symbols used today.



Berzelius' Symbols of Elements

Element Symbols:

- **Single Letters:** Elements like Carbon (C), Boron (B), and Oxygen (O) use their first letter.
- **Two Letters:** Elements like Aluminium (Al) and Chlorine (Cl) use the first letter and another in lowercase.

- **Latin Names:** Symbols like Iron (Fe) from ferrum and Sodium (Na) from natrium derive from Latin.

Atomic Mass

- **Atomic mass** is the mass of an atom of an element.
- The relative atomic mass is how many times an atom of an element is heavier than 1/12th of the mass of a carbon-12 atom.

Atomicity	NO. of Atoms	Examples
Monatomic	1	Helium (He), Neon (Ne), Argon (Ar)
Diatomic	2	Hydrogen (H ₂), Chlorine (Cl ₂), Nitrogen (N ₂)
Triatomic	3	Ozone (O ₃), Water (H ₂ O)
Tetratomic	4	Phosphorus (P ₄)
Polyatomic	more than 4	Sulfur (S ₈)

Molecule

- A molecule is the smallest particle of an element or compound that can exist independently.
- Examples: H₂O (water), O₂ (oxygen), O₃ (ozone).
- A single hydrogen atom (H) is not a molecule. When atoms bond, molecules such as H₂ (hydrogen gas) or H₂O (water) form.

Atomicity

The number of atoms constituting a molecule is referred to as atomicity.

Compound: A substance formed when two or more elements chemically combine. A compound has a fixed ratio of elements and distinct properties. Examples include Water (H₂O), Glucose (C₆H₁₂O₆), Calcium oxide (CaO), and Sodium chloride (NaCl).

Differences Between a Molecule and a Compound

- A **molecule** is formed when two or more atoms bond chemically and can consist of the same or different elements. While all compounds are molecules, not all molecules are compounds.
 - Example: Molecular hydrogen (H₂) is a molecule, not a compound. It consists of two hydrogen atoms.
- **Compound:** A molecule of two or more different elements bonded chemically.
 - Example: Water (H₂O) is a compound of hydrogen and oxygen atoms.

Ion

A charged particle is formed when an atom or molecule gains or loses electrons.

- **Cation:** A positively charged ion.
 - Examples: Na^+ , Ca^{2+} .
- **Anion:** A negatively charged ion.
 - Examples: F^- , Cl^- .

Chemical Formula

A **chemical formula** depicts the composition of a compound by using element symbols along with their corresponding valencies. Here's a brief overview:

- **Symbols of Elements:** Represent the elements in a compound.
- **Valency:** The combining capacity of an element, showing how atoms mix with others. For example, hydrogen (H^+) has a valency of 1, and oxygen (O^{2-}) has a valency of 2.

Rules for Writing Chemical Formulas:

1. **Balance Valencies:** Ensure that the total positive and negative charges are equal.
2. **Order of Elements:** Write the metal before the non-metal. For example, Sodium Chloride is NaCl .
3. **Polyatomic Ions:** If more than one polyatomic ion is present, place them in brackets. For example, calcium nitrate is written as $\text{Ca}(\text{NO}_3)_2$.

Writing Formulas for Simple Compounds (Binary Compounds):

1. Write the symbols of the elements.
2. Note the valencies of each component.

3. Crossover the valencies to balance the formula.

The formula for aluminium oxide:

Symbol Al O



Valency 3 2

Formula Al_2O_3

Molecular Mass

- **Definition:** The total of the atomic masses of all atoms in a molecule is called the **molecular mass**.
- **Calculation:** Multiply the atomic mass of each element by the number of its atoms in the molecule, then add these values.

Formula Unit Mass

- The sum of all atoms' atomic masses in an ionic compound's formula unit.
- **Calculation:** Add the atomic masses of the elements in the formula unit. For example, for Sodium Chloride (NaCl):

$$(1 \times 23) + (1 \times 35.5) = 58.5 \text{ u}$$

Mole Concept

- **Mole:** A unit representing 6.022×10^{23} entities (atoms, molecules, ions).
- **Avogadro's Number:** 6.022×10^{23} , used to calculate the number of particles in one mole of a substance.

Solved examples

1. Convert into mole. (a) 12 g of oxygen gas, (b) 20 g of water, (c) 22 g of carbon dioxide.

Solution: To convert grams to moles, use the formula:

$$\text{Moles} = \text{Mass (g)} / \text{Molar Mass (g/mol)}$$

(a) 12 g of Oxygen Gas (O_2)

- **Molar Mass of O_2 :** $2 \times 16 \text{ g/mol} = 32 \text{ g/mol}$
- **Moles of O_2 :** $12 \text{ g} / 32 \text{ g/mol} = 0.375 \text{ moles}$

(b) 20 g of Water (H_2O)

- **Molar Mass of H_2O :** $(2 \times 1) + 16 = 18 \text{ g/mol}$
- **Moles of H_2O :** $20 \text{ g} / 18 \text{ g/mol} \approx 1.11 \text{ moles}$

(c) 22 g of Carbon Dioxide (CO_2)

- **Molar Mass of CO_2 :** $12 + (2 \times 16) = 44 \text{ g/mol}$
- **Moles of CO_2 :** $22 \text{ g} / 44 \text{ g/mol} = 0.5 \text{ moles}$

2. Calculate the molar mass of the following substances. (a) Ethyne, C_2H_2 (b) Sulphur molecule, S_8 (c) Phosphorus molecule, P_4 (Atomic mass of phosphorus = 31) (d) Hydrochloric acid, HCl (e) Nitric acid, HNO_3

Solution:

To find the molar mass of a substance, the atomic masses of all the atoms present in one mole of the compound are added.

(a) Ethyne (C_2H_2)

- **Molar Mass:**

$$(2 \times \text{mass of C atom}) + (2 \times \text{mass of H atom})$$

$$= (2 \times 12) + (2 \times 1) = 24 + 2 = 26 \text{ g/mol}$$

(b) Sulphur Molecule (S_8)

- **Molar Mass:** $8 \times \text{Atomic mass of S}_8$
 $= 8 \times 32 = 256 \text{ g/mol}$

(c) Phosphorus Molecule (P_4)

- **Molar Mass:** $4 \times \text{Atomic mass of P}_4$
 $= 4 \times 31 = 124 \text{ g/mol}$

(d) Hydrochloric Acid (HCl)

- **Molar Mass:** Atomic mass of H
 $+ \text{Atomic mass of Cl}$
 $= 1 + 35.5 = 36.5 \text{ g/mol}$

(e) Nitric Acid (HNO_3)

- **Molar Mass:** (Atomic mass of H) +
 (Atomic mass of N) + (3 × Atomic mass of O)
 $= 1 + 14 + (3 \times 16) = 1 + 14 + 48 = 63 \text{ g/mol}$